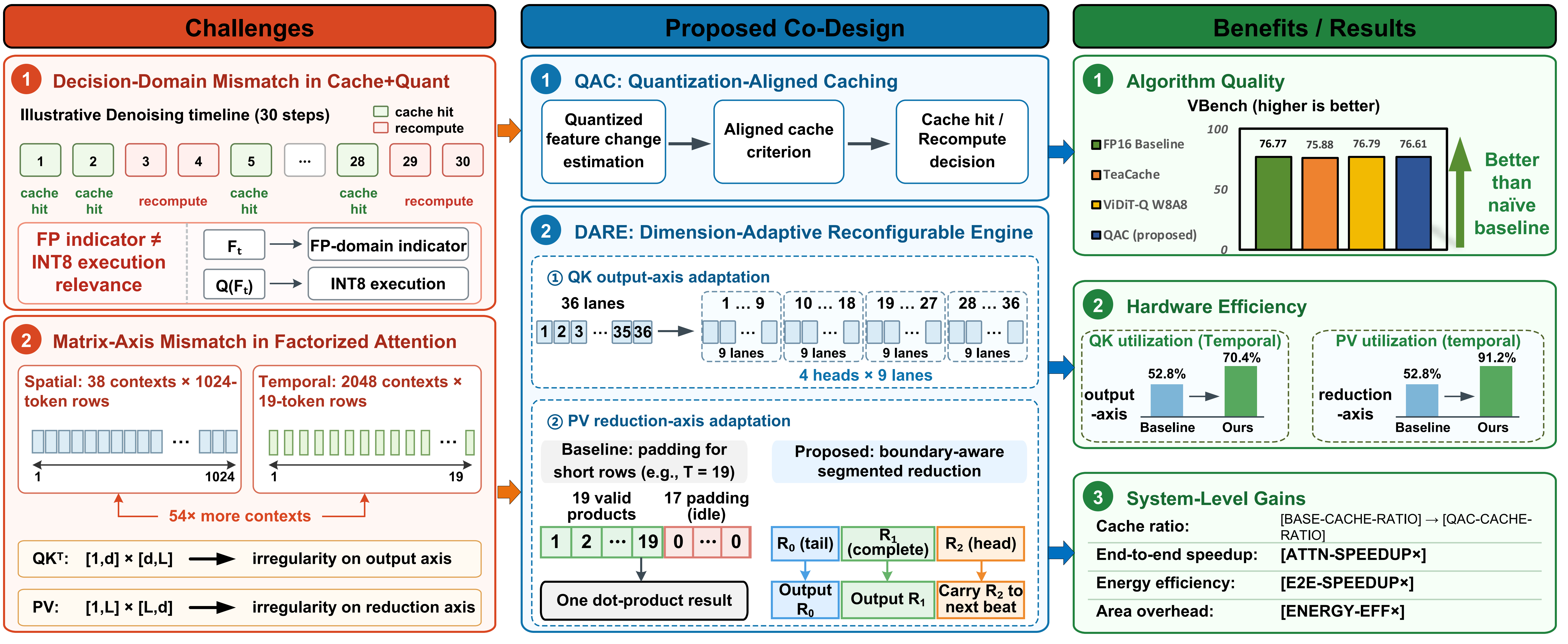
Challenges
Proposed Co-Design
Benefits / Results
1 Decision-Domain Mismatch in Cache+Quant
Illustrative Denoising timeline (30 steps)
cache hit
recompute
1 2 3 4 5 ··· 28 29 30
cache hit
cache hit
recompute
cache hit
cache hit
recompute
FP indicator ≠ INT8 execution relevance
F_t → FP-domain indicator
Q(F_t) → INT8 execution
2 Matrix-Axis Mismatch in Factorized Attention
Spatial: 38 contexts × 1024-token rows
1 1024
Temporal: 2048 contexts × 19-token rows
1 19
54× more contexts
QK^T: [1,d] × [d,L] irregularity on output axis
PV: [1,L] × [L,d] irregularity on reduction axis
1 QAC: Quantization-Aligned Caching
Quantized feature change estimation
Aligned cache criterion
Cache hit / Recompute decision
2 DARE: Dimension-Adaptive Reconfigurable Engine
① QK output-axis adaptation
36 lanes
1 2 3 ··· 35 36
1 … 9
10 … 18
19 … 27
28 … 36
9 lanes
4 heads × 9 lanes
② PV reduction-axis adaptation
Baseline: padding for short rows (e.g., T = 19)
19 valid products
17 padding (idle)
1 2 ··· 19 0 ··· 0
One dot-product result
Proposed: boundary-aware segmented reduction
R_0 (tail)
R_1 (complete)
R_2 (head)
Output R_0
Output R_1
Carry R_2 to next beat
1 Algorithm Quality
VBench (higher is better)
FP16 Baseline
TeaCache
ViDiT-Q W8A8
QAC (proposed)
100
50
0
76.77 75.88 76.79 76.61
Better than naïve baseline
2 Hardware Efficiency
QK utilization (Temporal)
output -axis
52.8%
70.4%
Baseline
Ours
PV utilization (temporal)
reduction -axis
52.8%
91.2%
Baseline
Ours
3 System-Level Gains
Cache ratio: [BASE-CACHE-RATIO] → [QAC-CACHE-RATIO]
End-to-end speedup: [ATTN-SPEEDUP×]
Energy efficiency: [E2E-SPEEDUP×]
Area overhead: [ENERGY-EFF×]